

TECHNICAL DOCUMENTATION: DRILL BIT ASSEMBLY

IDENTITY INFORMATION

Company_Name: Meridian Petroleum Operations LLC

Field_Engineer: James R. Mitchell

Field_Engineer_Id: MPO-ENG-4782

Document_Version: 2.3

Machine_Name: Offshore Platform Delta-7 Drilling System

Machine_Id: OPD7-DS-2024-001

Installation_Date: 2024-01-15

Number_Of_Units_In_Machine: 3

UNIT 1: PDC DRILL BIT - VERTICAL SECTION

Unit_Name: Polycrystalline Diamond Compact Bit (8.5" Vertical)

Unit_Id: PDC-V-8500-A1

Manufacture_Date: 2023-11-22

Manufacturer_Company_Name: TerraCore Drilling Solutions Inc.

Manufacturer_Company_Id: TCDS-MFG-0847

Inventory_Quantity: 2 units in stock

Supplier_Information: TerraCore Direct Supply, Houston TX | Lead Time: 14 days | Contact: procurement@terracore-drilling.com

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** 0–8,500 ft (vertical applications)
- **Operating_Pressure_PSI:** 2,800–3,200 PSI (Weight on Bit: 25,000–35,000 lbs)
- **Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds; pH 8.5–10.5
- **Known_Error_Codes:** ERR-PDC-001 (Cutter wear), ERR-PDC-002 (Bearing degradation), ERR-PDC-003 (Gauge loss)
- **Replacement_Part_Number:** PDC-V-8500-A1-REP
- **Maintenance_Schedule:** Inspect every 50 operating hours; replace cutters at 70% wear; full replacement at 500 operating hours

SECTION 1 - USE CASE

- Penetrates vertical wellbore sections in sandstone and shale formations

- Optimized for sustained drilling rates of 60–120 feet per hour
- Maintains gauge integrity and directional stability in vertical boreholes
- Designed for single-trip drilling operations up to 8,500 ft depth
- Reduces NPT (Non-Productive Time) through extended bit life and consistent ROP (Rate of Penetration)

SECTION 2 - FAILURE CAUSES

- **ERR-PDC-001 (Cutter Wear)** – Frequency: 45% | Caused by abrasive formation contact; accelerated by high WOB and low RPM
- **ERR-PDC-002 (Bearing Degradation)** – Frequency: 30% | Caused by inadequate mud cooling or contaminated drilling fluid
- **ERR-PDC-003 (Gauge Loss)** – Frequency: 15% | Caused by formation hardness exceeding design parameters or bit whirl
- **ERR-PDC-004 (Thermal Damage)** – Frequency: 7% | Caused by friction heat from low-flow mud circulation
- **ERR-PDC-005 (Junk Accumulation)** – Frequency: 3% | Caused by inadequate hole cleaning and debris in wellbore

SECTION 3 - SOLUTIONS

- **Cutter Wear Management:** Monitor ROP trends; reduce WOB if ROP drops >20%; implement API RP 54 bit grading protocol every 100 ft drilled
- **Bearing Protection:** Maintain mud flow rate ≥ 400 GPM; conduct mud analysis per OSHA 1910.1200 (hazard communication); replace bit if bearing temperature exceeds 180°F
- **Gauge Preservation:** Use offset drilling technique in hard formations; limit RPM to 120 in abrasive zones; inspect gauge diameter every 50 operating hours
- **Thermal Control:** Increase circulation rate by 15% in high-friction zones; monitor pump pressure; implement mud cooling system if surface temp $>140^{\circ}\text{F}$
- **Hole Cleaning:** Conduct wiper trips every 200 ft; maintain minimum annular velocity of 100 ft/min; use high-viscosity sweeps per API RP 13D

SECTION 4 - INVENTORY DATA

- **Current Stock:** 2 units (PDC-V-8500-A1) in warehouse; reorder threshold: 1 unit
- **Reorder Quantity:** 4 units per purchase order
- **Supplier:** TerraCore Drilling Solutions Inc. | Standard lead time: 14 days | Emergency expedite: 3 days (+15% cost)
- **Annual Consumption Rate:** 8–12 bits; estimated annual cost: \$48,000–\$72,000 USD

UNIT 2: ROLLER CONE BIT - DEVIATED SECTION

****Unit_Name:**** Three-Cone Roller Bit (6.125" Deviated)

****Unit_Id:**** RCB-D-6125-B2

****Manufacture_Date:**** 2023-10-18

****Manufacturer_Company_Name:**** PetroDrill Manufacturing Corp.

****Manufacturer_Company_Id:**** PDMC-MFG-1923

****Inventory_Quantity:**** 1 unit in stock

****Supplier_Information:**** PetroDrill Supply Chain, New Orleans LA | Lead Time: 21 days | Contact: sales@petrodrill-mfg.com

DOMAIN SPECIFICATIONS:

- ****Wellbore_Depth_Rating:**** 8,500–15,000 ft (deviated/build sections)
- ****Operating_Pressure_PSI:**** 2,400–2,900 PSI (Weight on Bit: 18,000–28,000 lbs)
- ****Drilling_Fluid_Compatibility:**** Water-based and synthetic muds; pH 8.0–10.0; avoid oil-based fluids
- ****Known_Error_Codes:**** ERR-RCB-001 (Cone bearing wear), ERR-RCB-002 (Shirttail erosion), ERR-RCB-003 (Seal failure)
- ****Replacement_Part_Number:**** RCB-D-6125-B2-REP
- ****Maintenance_Schedule:**** Inspect every 75 operating hours; bearing lubrication check every 150 hours; full replacement at 600 operating hours

SECTION 1 - USE CASE

- Drills deviated and build-section wellbores in mixed lithology (sandstone/limestone/shale)
- Provides controlled directional drilling with build rates of 2–4 degrees per 100 ft
- Maintains hole quality and minimizes torque variation in high-angle sections
- Supports extended-reach drilling (ERD) operations up to 15,000 ft measured depth
- Reduces differential sticking risk through optimized cone geometry and bearing design

SECTION 2 - FAILURE CAUSES

- ****ERR-RCB-001 (Cone Bearing Wear)**** – Frequency: 50% | Caused by inadequate lubrication or high rotational stress in build sections
- ****ERR-RCB-002 (Shirttail Erosion)**** – Frequency: 25% | Caused by high-velocity mud erosion and abrasive particle impact

- **ERR-RCB-003 (Seal Failure)** – Frequency: 15% | Caused by mud contamination or thermal cycling in deviated wells
- **ERR-RCB-004 (Cone Slippage)** – Frequency: 7% | Caused by bearing preload loss or inadequate WOB
- **ERR-RCB-005 (Gage Wear)** – Frequency: 3% | Caused by excessive side-load in high-angle sections

SECTION 3 - SOLUTIONS

- **Bearing Lubrication:** Implement API RP 54 bearing inspection protocol; inject bearing grease every 150 operating hours; replace bit if bearing torque increases >30%
- **Erosion Control:** Reduce mud velocity in shirrtail zone; use erosion-resistant mud additives per OSHA 1910.1200; monitor pump pressure for anomalies
- **Seal Integrity:** Conduct mud analysis per API RP 13D every 200 ft; maintain mud solids content <5%; replace bit if seal leakage detected
- **Cone Preload:** Verify WOB within design range (18,000–28,000 lbs); use rotary shouldered connections per API 7-1; inspect cone lock mechanism every 100 hours
- **Gage Management:** Limit build rate to 3°/100 ft in high-angle sections; reduce RPM to 80–100 in gage zone; conduct caliper survey every 500 ft

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 unit (RCB-D-6125-B2) available; reorder threshold: 1 unit
- **Reorder Quantity:** 3 units per purchase order
- **Supplier:** PetroDrill Manufacturing Corp. | Standard lead time: 21 days | Emergency expedite: 5 days (+20% cost)
- **Annual Consumption Rate:** 6–9 bits; estimated annual cost: \$54,000–\$81,000 USD

UNIT 3: HYBRID DRILL BIT - HORIZONTAL SECTION

Unit_Name: Hybrid PDC/Roller Bit (5.875" Horizontal)
Unit_Id: HYB-H-5875-C3
Manufacture_Date: 2023-12-05
Manufacturer_Company_Name: AdvancedBit Technologies Ltd.
Manufacturer_Company_Id: ABT-MFG-5641
Inventory_Quantity: 3 units in stock

****Supplier_Information:**** AdvancedBit Global Distribution, Aberdeen UK | Lead Time: 28 days |
Contact: logistics@advancedbit-tech.com

DOMAIN SPECIFICATIONS:

- ****Wellbore_Depth_Rating:**** 15,000–22,000 ft (horizontal/extended-reach sections)
- ****Operating_Pressure_PSI:**** 2,200–2,700 PSI (Weight on Bit: 15,000–22,000 lbs)
- ****Drilling_Fluid_Compatibility:**** Synthetic and oil-based muds; pH 7.5–9.5; compatible with invert emulsion fluids
- ****Known_Error_Codes:**** ERR-HYB-001 (PDC cutter fracture), ERR-HYB-002 (Roller cone spalling), ERR-HYB-003 (Hybrid interface separation)
- ****Replacement_Part_Number:**** HYB-H-5875-C3-REP
- ****Maintenance_Schedule:**** Inspect every 100 operating hours; interface bond check every 200 hours; full replacement at 700 operating hours

SECTION 1 - USE CASE

- Combines PDC cutting elements (center) with roller cones (gauge) for hybrid performance in horizontal wells
- Optimized for long-radius and extended-reach drilling with build rates $<1^{\circ}/100$ ft
- Maintains directional stability and hole quality in high-angle ($>70^{\circ}$) and horizontal (90°) sections
- Achieves drilling rates of 40–90 feet per hour in mixed lithology formations
- Reduces bit trips and improves economics in deepwater and onshore extended-reach applications

SECTION 2 - FAILURE CAUSES

- ****ERR-HYB-001 (PDC Cutter Fracture)**** – Frequency: 40% | Caused by impact loading in hard streaks or inadequate WOB distribution
- ****ERR-HYB-002 (Roller Cone Spalling)**** – Frequency: 35% | Caused by bearing fatigue in horizontal sections with high side-load
- ****ERR-HYB-003 (Hybrid Interface Separation)**** – Frequency: 15% | Caused by thermal cycling or manufacturing defect in PDC-to-cone bond
- ****ERR-HYB-004 (Mud Ring Erosion)**** – Frequency: 7% | Caused by high-velocity mud circulation in horizontal sections
- ****ERR-HYB-005 (Stabilizer Wear)**** – Frequency: 3% | Caused by friction against wellbore wall in high-angle sections

SECTION 3 - SOLUTIONS

- ****PDC Cutter Protection:**** Maintain WOB within 15,000–22,000 lbs range; reduce RPM to 60–80 in hard formations; implement API RP 54 cutter inspection every 100 hours

- **Cone Bearing Management:** Monitor bearing temperature continuously; limit side-load by maintaining vertical well alignment; replace bit if bearing preload loss exceeds 10%
- **Interface Bond Integrity:** Conduct ultrasonic inspection per manufacturer protocol every 200 operating hours; avoid thermal shock (rapid mud temperature changes); replace bit if separation detected
- **Mud Ring Protection:** Reduce circulation rate by 10% in horizontal sections; use erosion-resistant mud additives; monitor pump pressure for sudden increases
- **Stabilizer Maintenance:** Inspect stabilizer blades every 100 hours; replace if wear exceeds 0.125"; use low-friction drilling fluid to reduce friction torque

SECTION 4 - INVENTORY DATA

- **Current Stock:** 3 units (HYB-H-5875-C3) in warehouse; reorder threshold: 2 units
- **Reorder Quantity:** 5 units per purchase order
- **Supplier:** AdvancedBit Technologies Ltd. | Standard lead time: 28 days | Emergency expedite: 7 days (+25% cost)
- **Annual Consumption Rate:** 10–14 bits; estimated annual cost: \$85,000–\$119,000 USD

COMPLIANCE SUMMARY

OSHA 1910 Standards Applied:

- 1910.1200 (Hazard Communication) – Drilling fluid compatibility and mud analysis requirements
- 1910.212 (General Machine Guarding) – Bit handling and installation safety procedures
- 1910.268 (Telecommunications) – Communication protocols for bit condition monitoring

API Standards Applied:

- API RP 54 (Recommended Practice for Oilwell Drilling Fluids) – Mud compatibility and maintenance schedules
- API RP 13D (Recommended Practice for the Rheology and Hydraulics of Oil-Gas Drilling Fluids) – Hole cleaning and circulation parameters
- API 7-1 (Specification for Rotary Shouldered Connections) – Bit connection integrity and torque specifications

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Authorized By: Regional Operations Manager, Meridian Petroleum Operations LLC

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